

Analysis and Partial Results on the Erdős Conjecture on Arithmetic Progressions

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Abstract

This document provides a rigorous axiomatic decomposition, literature review, and step-by-step partial proof structure for the Erdős Conjecture on Arithmetic Progressions. The architecture is designed to facilitate future autoformalization in systems such as Lean 4.

1 Analysis and Decomposition

1.1 Axiomatic Definitions

Let $\mathbb{N} = \{1, 2, \dots\}$ denote the set of positive integers. We define the types and variables rigorously.

Definition 1 (Divergent Harmonic Sum). *Let $A \subseteq \mathbb{N}$ be a subset. We say that A has a divergent harmonic sum, denoted as $\mathcal{D}(A)$, if the series of reciprocals of its elements diverges:*

$$\sum_{n \in A} \frac{1}{n} = \infty.$$

Definition 2 (Arithmetic Progression of Length k). *For $k \in \mathbb{N}, k \geq 3$, a set $A \subseteq \mathbb{N}$ contains an arithmetic progression of length k , denoted as $\mathcal{AP}_k(A)$, if there exist $a \in \mathbb{N}$ and $d \in \mathbb{N}$ such that for all $0 \leq i < k$, $a + id \in A$.*

Conjecture Statement: For any $A \subseteq \mathbb{N}$, if $\mathcal{D}(A)$ holds, then for all $k \geq 3$, $\mathcal{AP}_k(A)$ holds.

1.2 Underlying Algebraic and Combinatorial Structures

The problem fundamentally intertwines additive combinatorics and analytic number theory. The set A is viewed as a dense subset in a weighted sense. The natural structure involves Fourier analysis on the cyclic groups $\mathbb{Z}/N\mathbb{Z}$ (the Hardy-Littlewood circle method) and the use of Gowers norms to detect linear structures. An alternative perspective uses ergodic theory, mapping the subset A to a measure-preserving dynamical system.

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2 Contextual Literature Research

The most prominent theorem related to this conjecture is Szemerédi's Theorem, which establishes that any subset of $\mathbb{N}$ with positive upper density contains arbitrarily long arithmetic progressions. The Green-Tao theorem provides a massive breakthrough by proving that the primes, a set with vanishing density but divergent harmonic sum, contain arbitrarily long arithmetic progressions. Recent advancements by Kelley and Meka (2023) on Roth's theorem give strong quantitative bounds on sets without 3-term arithmetic progressions.

By analogy, the resolution of the Cap Set Problem by Ellenberg and Gijswijt utilizes the polynomial method. While finite field methods do not directly translate to $\mathbb{Z}$, algebraic methods analogous to Croot-Lev-Pach bounds provide insights into bounding subsets without linear structures.

3 Proof Strategy & Isolation of Lemmas

To tackle a simplified version (e.g., $k = 3$ over restricted subsets), we decompose the problem into several lemmas.

- **Lemma 1 (Density Increment):** If a subset A of $\{1, \dots, N\}$ lacks 3-term arithmetic progressions, it must correlate with a Bohr set, allowing an increment in density on a structured substructure.
- **Lemma 2 (Harmonic Bound on Sparsity):** If a set A contains no 3-term arithmetic progressions, its counting function $A(x) = |A \cap \{1, \dots, x\}|$ is bounded by $O(x/(\log x)^c)$ for some $c > 1$.

4 Informal Proof of Lemma 2 (Zero Ellipsis)

Proof of Lemma 2. We present the proof step by step. Let $A \subset \mathbb{N}$ such that A contains no 3-term arithmetic progressions. Let N be a large integer, and let $A_N = A \cap \{1, 2, \dots, N\}$. According to the quantitative bounds by Kelley and Meka, the size of A_N satisfies the inequality:

$$|A_N| \leq \frac{N}{\exp(c(\log N)^\beta)}$$

for some constants $c > 0$ and $\beta > 0$. We evaluate the harmonic sum of A by partial summation. Let $S(N) = \sum_{n \in A, n \leq N} \frac{1}{n}$. Using summation by parts, we express $S(N)$ in terms of $A(x) = |A \cap \{1, \dots, \lfloor x \rfloor\}|$:

$$S(N) = \frac{A(N)}{N} + \int_1^N \frac{A(t)}{t^2} dt.$$

We bound the first term:

$$\frac{A(N)}{N} \leq \frac{1}{\exp(c(\log N)^\beta)}.$$

As N approaches infinity, this term approaches 0. For the integral term, we substitute the bound for $A(t)$:

$$\int_1^N \frac{A(t)}{t^2} dt \leq \int_1^N \frac{t \exp(-c(\log t)^\beta)}{t^2} dt = \int_1^N \frac{1}{t \exp(c(\log t)^\beta)} dt.$$

We compute this integral using the substitution $u = \log t$. Thus, $du = \frac{1}{t}dt$. The limits of integration change from $t = 1$ to $u = 0$, and $t = N$ to $u = \log N$:

$$\int_0^{\log N} \frac{1}{\exp(cu^\beta)} du.$$

For any $\beta > 0$, the integral $\int_0^\infty \exp(-cu^\beta) du$ is convergent. Specifically, letting $v = cu^\beta$, $dv = c\beta u^{\beta-1} du$, we find that the integral is proportional to the Gamma function $\Gamma(1/\beta)$. Since the integral is bounded by a finite constant independent of N , it follows that $S(N)$ is bounded as $N \rightarrow \infty$. Thus, $\sum_{n \in A} \frac{1}{n} < \infty$. By contraposition, if $\sum_{n \in A} \frac{1}{n} = \infty$, the set A must contain a 3-term arithmetic progression. This completes the proof of Lemma 2. $\square$

5 Architecture for Autoformalization

The proof structure is directly mapped for implementation in Lean 4.

```
import Mathlib.Data.Real.Basic
import Mathlib.Analysis.SpecialFunctions.Log.Basic

-- Axiomatic definition of the divergent harmonic sum
def has_divergent_harmonic_sum (A : Set Nat) : Prop :=
  Filter.Tendsto (fun N => \sum_{n \in A \cap {x | x \le N}} (1 / (n : Real)))
    Filter.atTop Filter.atTop

-- Axiomatic definition of containing an arithmetic progression of length k
def contains_AP (A : Set Nat) (k : Nat) : Prop :=
  \exists a d : Nat, d > 0 \and \forall i : Nat, i < k \to a + i * d \in A

-- Main Theorem Statement (Erdős Conjecture)
theorem erdos_ap_conjecture (A : Set Nat) (h : has_divergent_harmonic_sum A) :
  \forall k \ge 3, contains_AP A k := by
  sorry

-- Lemma 2
lemma ap3_harmonic_bound (A : Set Nat) (h : \not contains_AP A 3) :
  \not has_divergent_harmonic_sum A := by
  sorry
```